PS Description:

Anomaly detection in hyperspectral and thermal remote sensing datasets is crucial for various applications, including environmental monitoring, mining, precision agriculture, and disaster management. These datasets provide rich spectral and/or thermal information, enabling the identification of materials, objects, or behaviors that deviate from the norm.

Hyperspectral data cubes, with their hundreds of contiguous spectral bands, allow for material distinction based on unique spectral signatures. Deep learning-based spectral-spatial anomaly detection can enhance data interpretability and material identification by suppressing natural anomalies.

Similarly, deep learning can be used to detect subtle temperature changes in thermal datasets, such as those caused by industrial operations, fire incidents, or machine operations, facilitating subsequent material identification through thermal emissivity referencing.

Objectives:

1) Manmade anomaly detection using satellite hyperspectral images (400-2500 nm) through deep learning and Generative AI.
2) Characterization of materials in detected anomalies (pixels) through spectral data analytics.
3) Thermal anomalies detection associated with human activities using deep learning techniques.
4) Characterization and identification of thermal anomalies using thermal emissivity of various materials.
5) Multimodal sensor data fusion (thermal and Hyperspectral) to augment anomaly detection.

Datasets used for training:

Dataset	Sensor	Spectral range(nm)	Spatial Size	Bands	Res. (m)	Background	Anomalies
San Diego	AVIRIS	370-2510	100*100	189	3.5	Airport	Airplanes
HYDICE	HYDICE	400-2500	80*100	175	1.56	Urban area	Roofs, cars
Salinas	AVIRIS	400-2500	120*120	204	3.7	Salinas Valley	Rooftops

Approach:

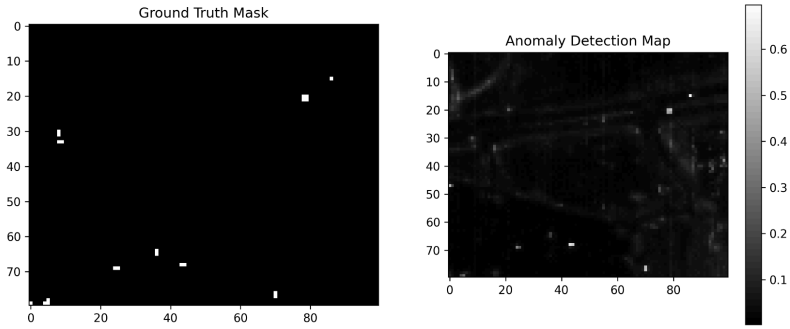
The **core idea** of the models used until now is instead of trying to find what's anomalous, we train a network which gets really good at **reconstructing only the background** of the image. They assume **anomalies are rare** and different from the background. When this trained network tries to process the original image, it successfully rebuilds the background parts but **fails to rebuild the anomalous parts**.

1) ***Hyperspectral anomaly detection generative adversarial network (HADGAN)-***

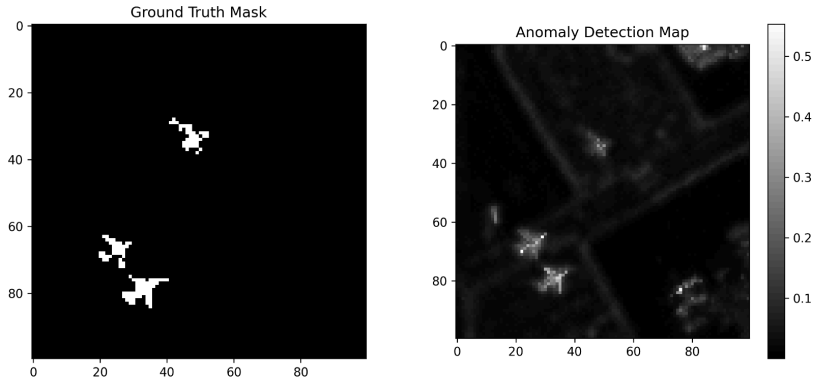
The HADGAN model has two main parts:

- **Discriminative Reconstruction Network:** It's built on an **Autoencoder (AE)**. They modify this AE with several components from **Generative Adversarial Networks (GANs)** to force it to learn *only* the background distribution:
  - **Adversarial Latent Discriminator (DZ)**: This component forces the latent representation of the data to conform to a normal Gaussian distribution, which is assumed to represent the background.
  - **Adversarial Image Corrector (DI)**: A second discriminator is used to ensure that the reconstructed background image is realistic and doesn't contain strange artifacts that GANs can sometimes produce.
- **Joint Spatial-Spectral Detector:** Once the residual image (with highlighted anomalies) is generated, they apply a final detection step that looks at both spatial and spectral properties:
  - **Energy-based Spatial Detection**: It finds anomalies based on their physical attributes like shape and size within the image. They propose a novel "minimum energy" method to select the most informative bands for this step.
  - **Distance-based Spectral Detection**: This part identifies anomalous pixels based on how different their spectral signature (i.e., pixel value in the residual image) is from the norm, using a method similar to the classic RX detector.

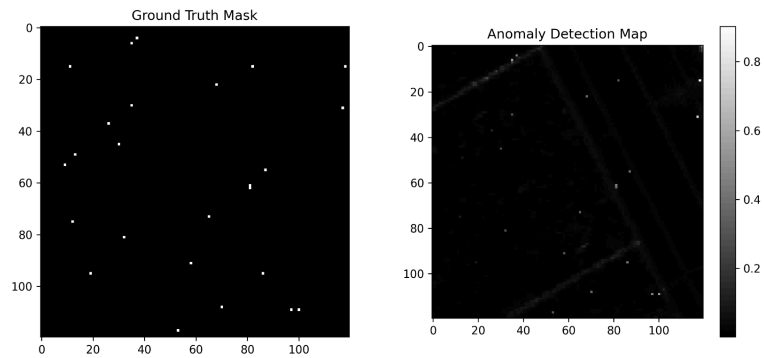
Results:



For HYDICE Urban dataset



For San Diego dataset



For Salinas dataset

These are continuous anomaly maps and not binary thresholded maps

References:

Discriminative_Reconstruction_Constrained_Generative_Adversarial_Netw…

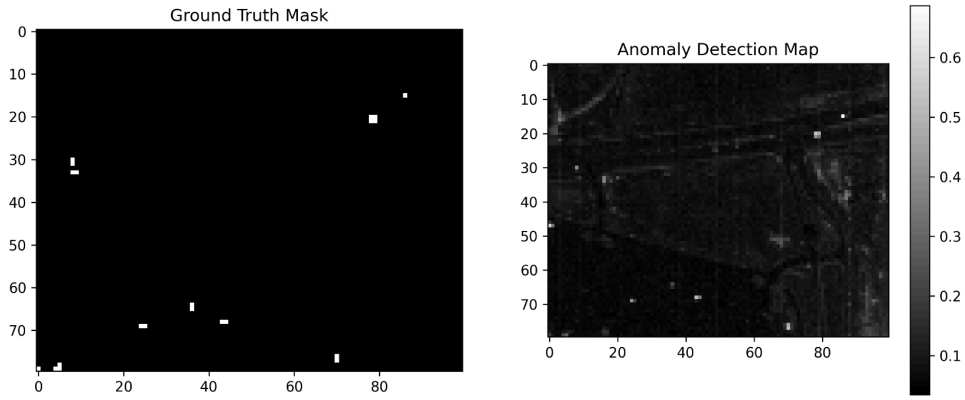
2) *Robust graph autoencoder (RGAE)-*

The RGAE model introduces two key innovations:

1. Robustness to Anomalies: A standard AE can be "too good" at learning and might reconstruct anomalous pixels well if they are in the training data, making them harder to detect. To fix this, they replace the standard reconstruction loss function with an
   l2,1-norm. This simple change normalizes the error gradients during training, which reduces the influence of pixels with large errors (the anomalies) and forces the network to focus on learning the patterns of the majority data (the background). This makes the reconstruction process more
   robust to contamination from outliers.

2. Efficient Spatial Structure Preservation: AEs typically ignore the spatial relationship between pixels. While graph-based methods can preserve this structure, they are often extremely slow because they require comparing every pixel to many others. The authors propose
"SuperGraph," a much faster method:
    - First, the image is segmented into superpixels—small, homogenous regions where pixels are spatially and spectrally similar.
    - Then, the graph that captures pixel relationships is built only within each superpixel.
    - This massively reduces computation time while still effectively incorporating local spatial and geometric structure into the model.

Results:



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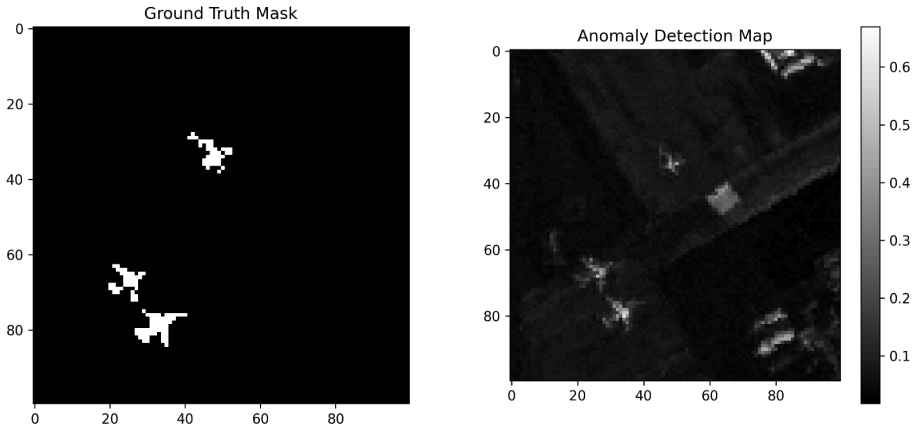
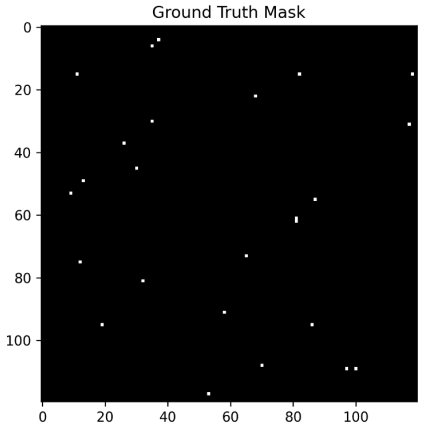
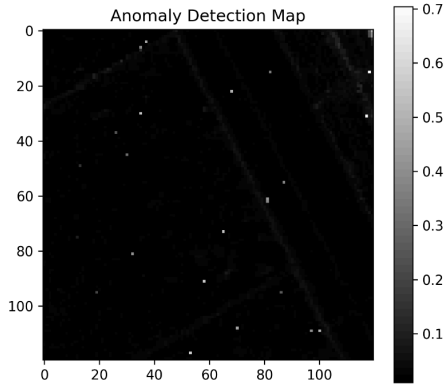


For San Diego dataset





For Salinas dataset

References:

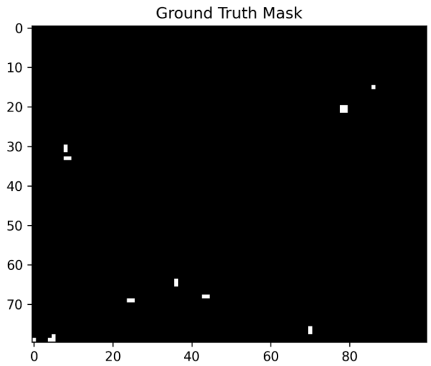
Hyperspectral_Anomaly_Detection_With_Robust_Graph_Autoencoders.pdf

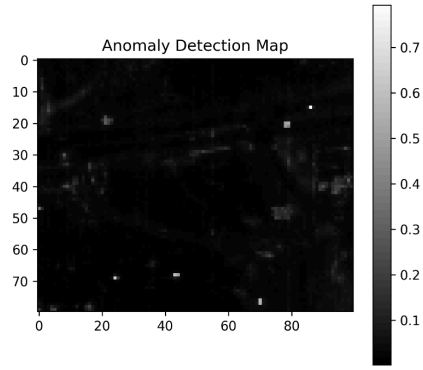
Robust_Graph_Autoencoder_for_Hyperspectral_Anomaly_Detection.pdf

3) HADGAN+RGAE-

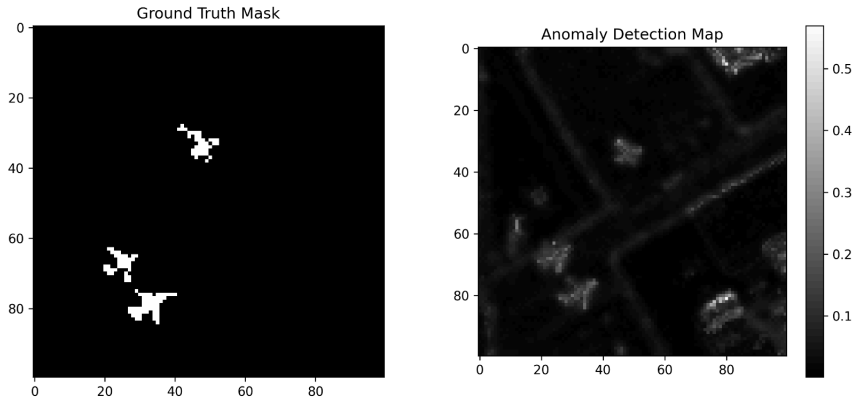
Now I combined this HADGAN and RGAE architectures in a way that uses the AE structure and postprocessing step from HADGAN but uses the supergraph and l21 norm of RGAE.

Results:

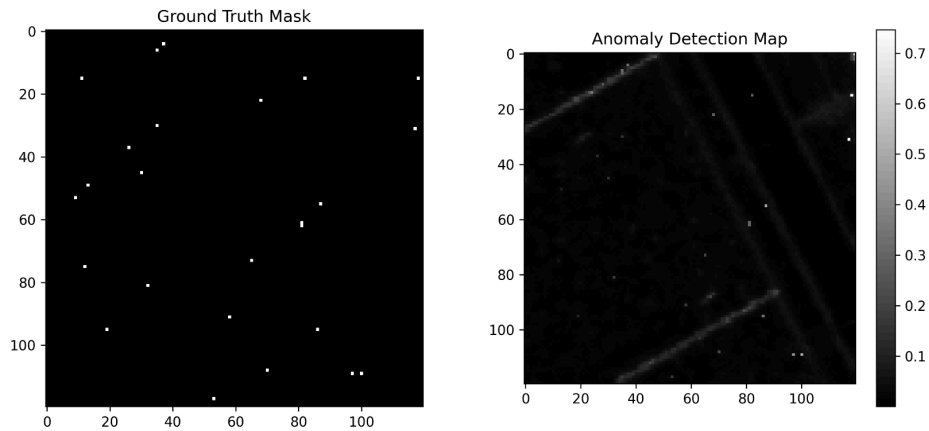




For HYDICE Urban dataset



For San Diego dataset



For Salinas dataset

f1

Dataset	HADGAN+RGAE	HADGAN	RGAE
SanDiego	0.42660550458715596	0.6007604562737643	0.38155136268343814
Salinas	0.5833333333333334	0.8181818181818182	0.8333333333333334
HYDICE-urban	0.6451612903225806	0.6470588235294118	0.6060606060606061

pr-auc

Dataset	HADGAN+RGAE	HADGAN	RGAE
SanDiego	0.25218597462258485	0.5901442765444482	0.22905387241953934
Salinas	0.5331973374946276	0.7702548695688198	0.8285333008505753
HYDICE-urban	0.6455427136474703	0.6346307195555387	0.5226418642837863

roc-auc

Dataset	HADGAN+RGAE	HADGAN	RGAE
SanDiego	0.9659897855139467	0.9894897597961944	0.9561270275421998
Salinas	0.9794838260869566	0.9922059130434783	0.9971812173913044
HYDICE-urban	0.9941572819126396	0.9943124511366146	0.9829552575510716

Considering the metrics above, I tried to modify the HADGAN model.

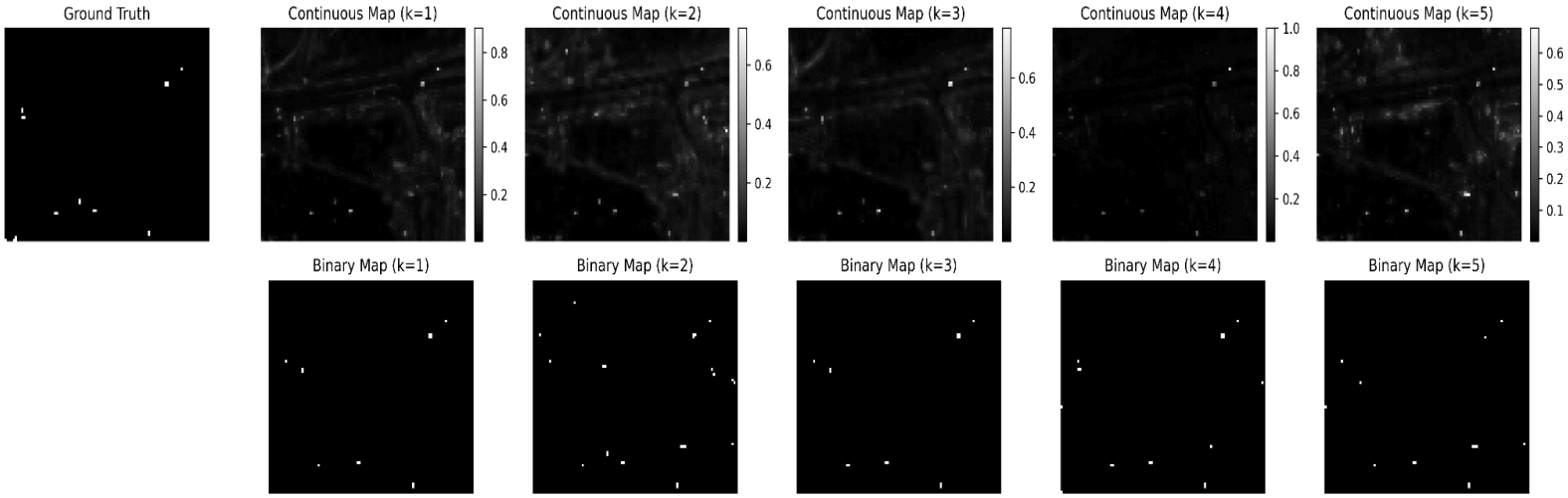
To further enhance the performance of the HADGAN model, I implemented an improved training strategy inspired by the original Generative Adversarial Network (GAN) paper.
Generative Adversarial Nets.pdf

The core idea of a GAN is a minimax game between a generator and a discriminator. The quality of the generator is highly dependent on the quality of the gradients it receives from the discriminator. A weak or under-trained discriminator may provide misleading feedback, slowing down the generator's learning process.

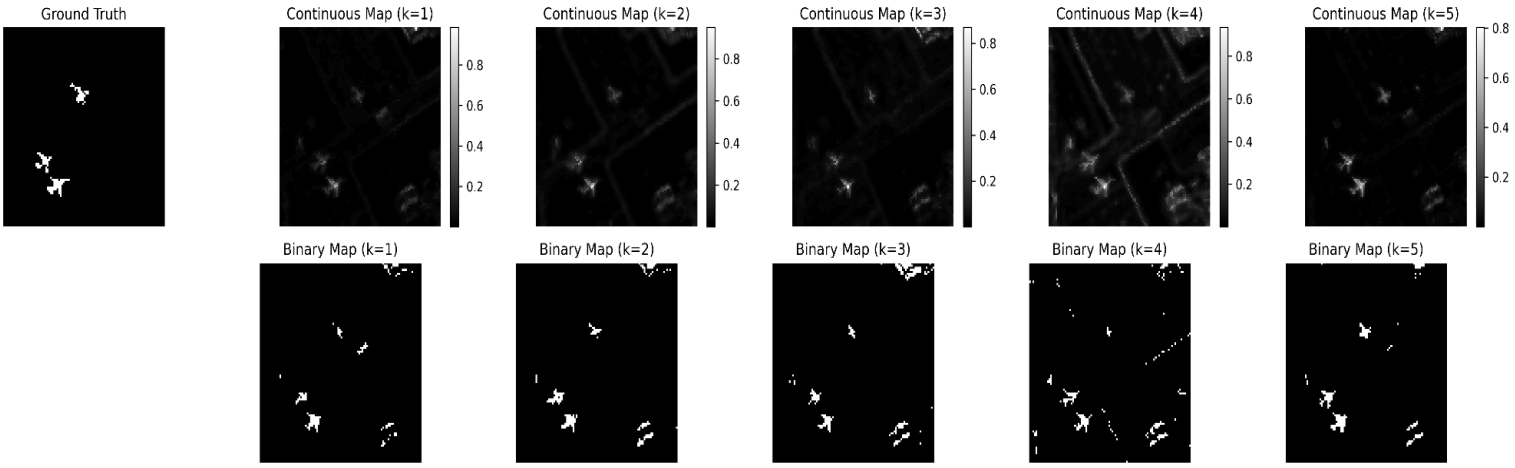
The original GAN paper suggests that to maintain a strong adversary, one can perform **k** update steps for the discriminator for every one update step of the generator. This approach helps ensure the discriminator remains near its optimal solution for the current generator, thereby providing more stable and potent gradients.

I have adapted this principle to the HADGAN framework. The training loop was modified to update the model's discriminators—both the **latent discriminator (DZ)** and the **image discriminator (DI)**—for 'k' iterations for every single training iteration of the autoencoder (the generator).
By allowing the discriminators to become more proficient before the generator is updated, we compel the autoencoder to produce a more accurate and faithful reconstruction of the background.

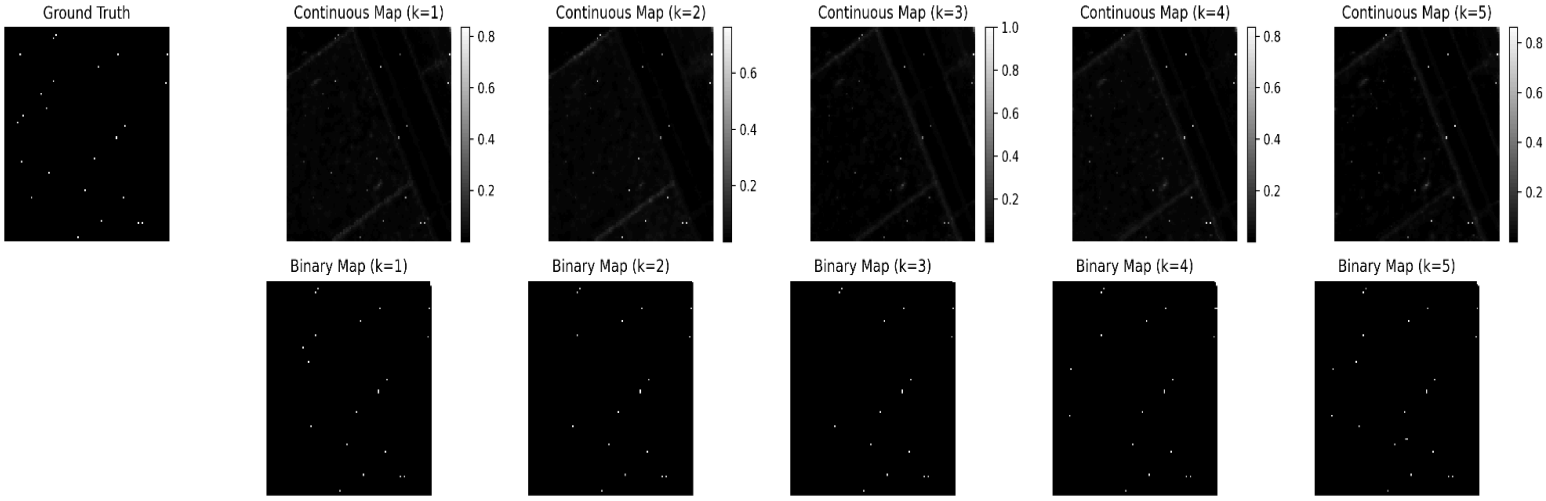
<u>Results:</u> 1) HADGAN with k step Discriminator update



For Hydice urban dataset

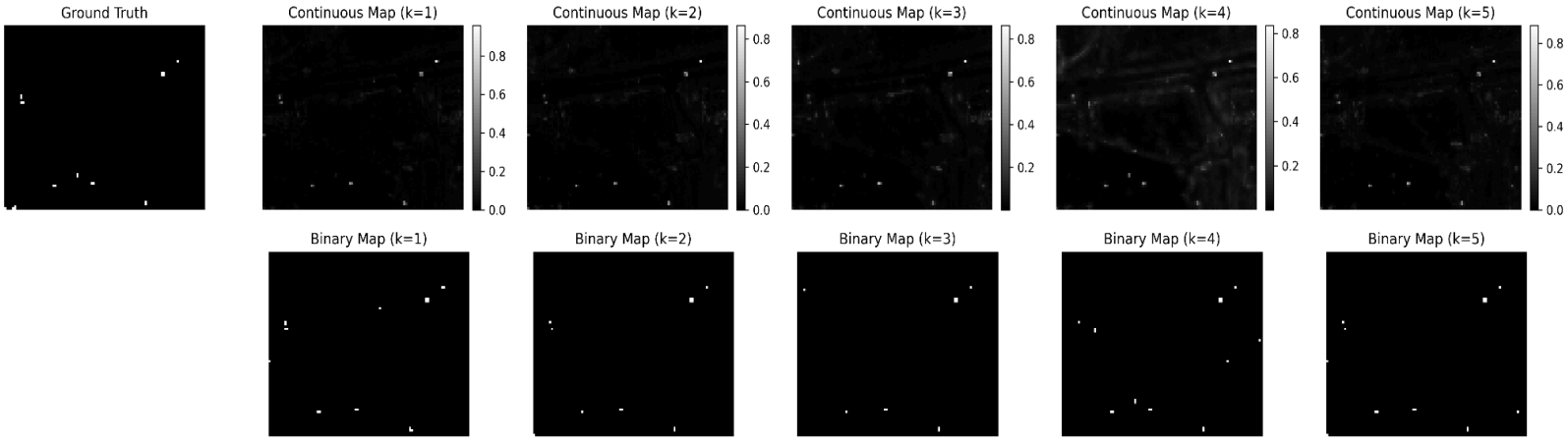


For San Diego dataset

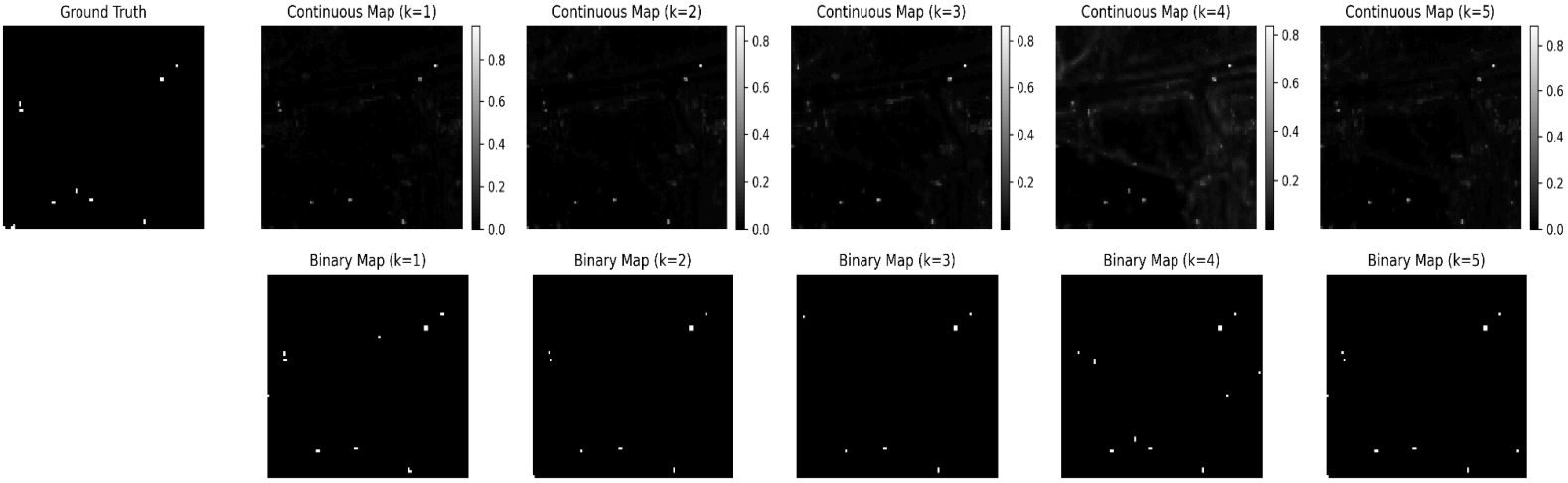


For Salinas dataset

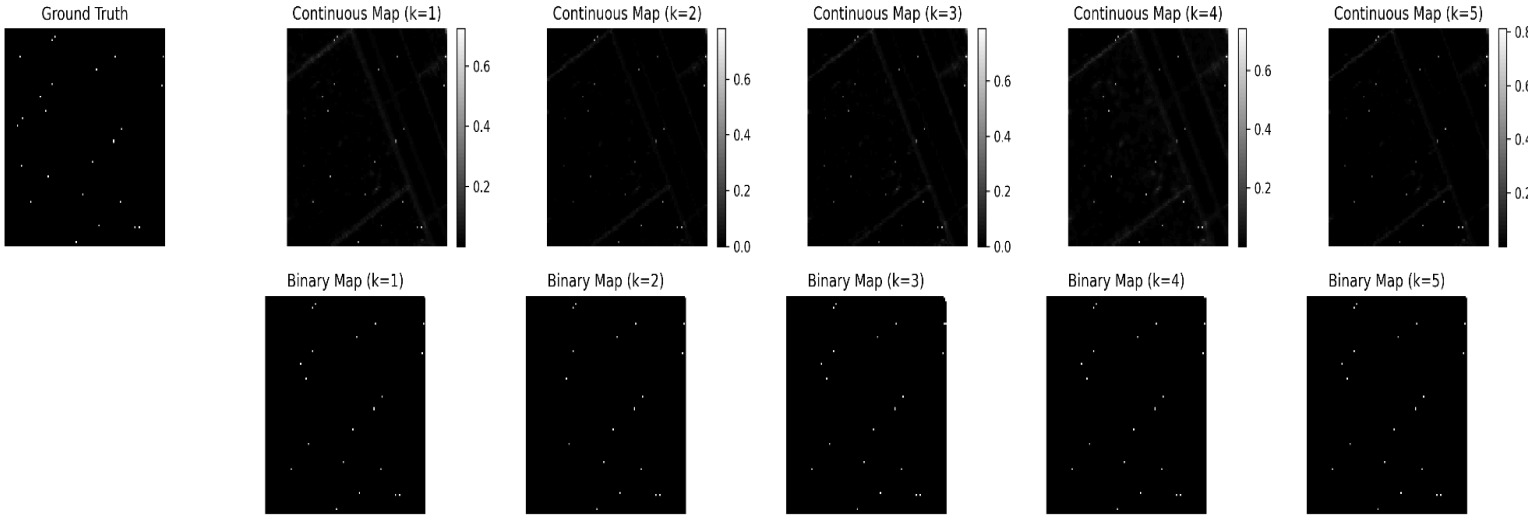
2) HADGAN with k step Discriminator update and l21 norm



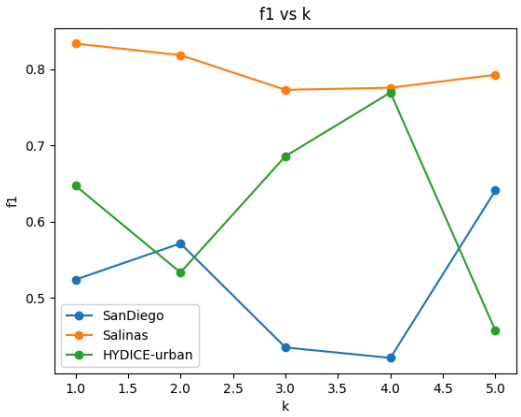
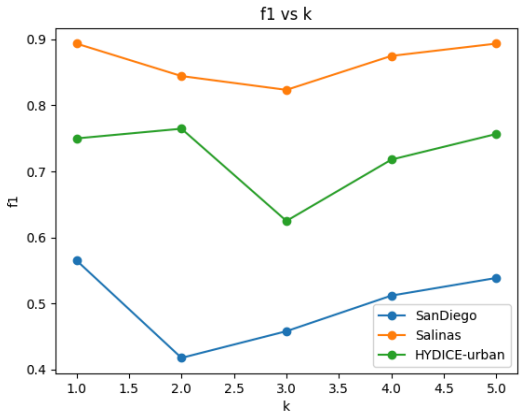
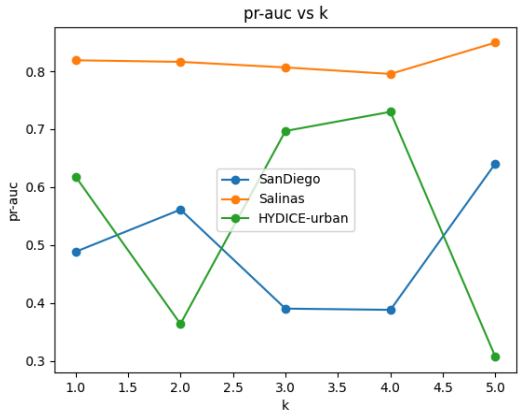
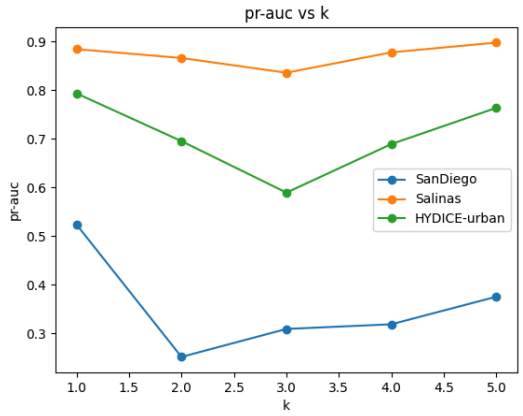
For HYDICE urban dataset

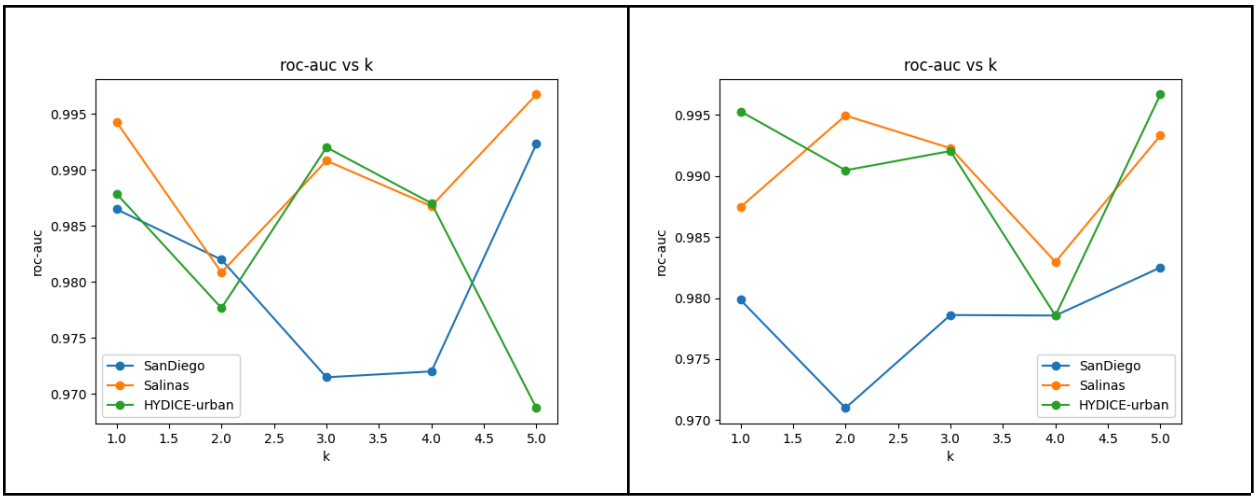


For San Diego dataset



For Salinas dataset

HADGAN (K step Discriminator update)	HADGAN (K step Discriminator update with l21 norm)
	
	


roc-auc vs k
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